WORKSHEET

Tossing 36 coins simultaneously

If you toss 36 coins at the same time, over and over again, and count the number of heads, you will find that the results are normally distributed with a mean of 18 heads and a standard deviation of 3. We will investigate the claim that about $\frac{2}{3}$ of scores are within one standard deviation of the mean and that it is extremely unlikely to obtain a score that is more than three standard deviations from the mean.

Range of scores one standard deviation from the mean $= (18 - 3)$ to $(18 + 3)$
$= 15$ to 21

Range of scores three standard deviations from the mean $= (18 - 3 \times 3)$ to $(18 + 3 \times 3)$
$= 9$ to 27

Download the spreadsheet 'Tossing 36 coins simultaneously' from www.nelsonnet.com.au. The spreadsheet will simulate tossing 36 coins simultaneously and will count the number of heads each time.

1 Work in pairs to complete this class activity. Decide who will run the simulations on the spreadsheet and who will record the results on the record sheet (see the next page). Change roles part of the way through the experiment.

2 Complete 40 simulations and record the results of each trial on the sheet, then complete the totals at the end.

3 Combine the data each pair collected, to make a class summary.

4 Calculate the percentage of times that 15 to 21 heads came up.

5 Calculate the percentage of times that the number of heads was 8 or less, or 28 or more.

6 Did approximately $\frac{2}{3}$ $(66\frac{2}{3}\%)$ of the scores lie within one standard deviation of the mean? _____

7 Were there many scores further than three standard deviations from the mean? _____

8 The heights of Australian women are normally distributed with a mean of 162 cm and a standard deviation of 11 cm. Australian Opals basketballer Elizabeth Cambage has a height of 203 cm, which is 3.7 standard deviations above the mean. Explain what this means by comparing Elizabeth's height to the heights of Australian women.

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RECORD SHEET FOR TOSSING 36 COINS SIMULTANEOUSLY

Pairs data

Number of heads	Tally	Frequency
0–8		
9–14		
15–21		
22–27		
28–36		
	Total:	**40**

Class summary

Number of heads	Frequency
0–8	
15–21	
28–36	
Total:	

Calculations

1 Percentage of scores within one standard deviation of the mean (15 to 21 heads) $= \dfrac{\text{number of times there were 15 to 21 heads}}{\text{total number of simulations}} \times 100\%$

$= \dfrac{\qquad\qquad}{\qquad\qquad}$

2 Percentage of scores beyond three standard deviations of the mean (8 or less, or 28 or more heads) $= \dfrac{\qquad\qquad}{\qquad\qquad}$

$= \dfrac{\qquad\qquad}{\qquad\qquad}$

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